Derivative Placement in Distributed Nonlinear Finite Elements: Equivalence, Verification, and Evidence

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Abstract

Second derivatives of nonlinear finite-element energies may be constructed at the element, constitutive, or assembled sparse-operator level. Element and constitutive differentiation are mathematically equivalent when they differentiate the same quadrature functional, affine constraint map, and smooth constitutive branch. Colored recovery is guaranteed to give the same sparse operator under additional complete-pattern, interference-coloring, overlap, and row-ownership conditions. We state these two conditional results and implement the three routes with JAX and PETSc.

The verification separates derivative correctness from nonlinear-solver and timing evidence. In prerelease diagnostic executions, fixed-element tests with linear and high-order tetrahedra, element and constitutive differentiation agree to relative errors below 2.4×10^{-16}. Material-point tests exercise all five branches of a three-dimensional Mohr–Coulomb endpoint surrogate, while explicitly excluding ordinary Hessian claims at switches. A one-versus-two-rank hyperelastic test reproduces bitwise-identical stored residual and matrix arrays and the tangent action to 2.3×10^{-16}. Independent manufactured problems recover the expected P1 rates for p-Laplace and Ginzburg–Landau equations. A nonaffine hyperelastic problem gives last-pair orders 1.887, 1.006, and 0.983 for displacement, deformation gradient, and first Piola stress, respectively.

An enriched-quadrature diagnostic also shows that nearly unchanged energy can coexist with materially different residuals and tangent actions. The present diagnostics therefore exercise the conditional derivative identities and the fail-closed comparison protocol, but they are not clean-release evidence and do not support a universal route ranking or timing claim. The accompanying implementation provides the tests, canonical state exports, analysis contracts, and a prespecified distributed crossover protocol.

1 Introduction

Nonlinear finite-element models are often defined by compact energy or constitutive programs, while their solution requires distributed sparse matrices, Krylov methods, and nonlinear globalization. Automatic differentiation (AD) removes many manual derivative formulas, but it does not remove the choice of where differentiation occurs. A tangent may be obtained from the complete element energy, from a low-dimensional constitutive density, or from colored actions of the assembled Hessian. Each route exposes a different combination of local arithmetic, memory traffic, sparse insertion, and communication.

This choice is scientific only if the compared routes represent the same discrete problem. Quadrature, essential constraints, affine lifts, active constitutive branches, engineering-shear conventions, and state ordering must remain fixed. Distributed colored recovery introduces further requirements: the structural pattern must be complete, same-color columns must not interfere in an owned row, and every ghost value needed by a local action must be available. Agreement of endpoint energies alone does not verify these conditions.

Existing frameworks establish the principal component technologies. High-level finite-element systems automate variational forms and parallel assembly [1, 2]; JAX-based methods provide differentiable mechanics and inverse problems [3, 4]; and bridge architectures combine differentiable local programs with established sparse solvers [5–7]. Recent work also develops locality-aware element differentiation, global AD with sparse tangent materialization, and automatic sparsity detection [8–10]. Consequently, the novelty of the present study is not JAX–PETSc coupling, constitutive AD, or graph coloring in isolation.

We instead ask three narrower questions. First, which assumptions guarantee that element AD, constitutive AD, and colored sparse recovery differentiate one quadrature-defined potential? Second, which fixed-state tests are designed to expose errors in branch selection, local contraction, sparse recovery, or MPI ownership? Third, what evidence is required before descriptive route timings support a crossover or selection claim? The third question is treated as an experimental protocol in this revision because the required paired distributed measurements are not yet available.

1.1 Contributions

The paper makes four contributions.

1. We formulate a branchwise equivalence result for element and constitutive differentiation of a fixed quadrature potential. The statement includes the constrained affine map and excludes branch switches and unresolved repeated spectra.
2. We state the interference, overlap, and row-ownership conditions under which colored Hessian actions recover the same distributed sparse operator.
3. We provide an evidence hierarchy that separates analytic and manufactured verification, canonical fixed-state MPI checks, nonlinear stopping, and equal-accuracy timing. Failed admission gates prevent a route selector or speed claim from being fitted.
4. We exercise this hierarchy on smooth scalar energies, orientation-preserving hyperelasticity, and a synthetic branch-structured Mohr–Coulomb endpoint potential. The latter is used to test derivative placement and is not presented as incremental plasticity validation.

The current diagnostic results are deliberately asymmetric. The correctness and manufactured-solution blocks pass for the tested local states, and a canonical hyperelastic state is reproduced across one and two MPI ranks. These rows still require clean-release reproduction before submission. The available route timings, however, do not satisfy the clean, paired, train–holdout design required for a predictive cost model. We therefore report fixed-state correctness diagnostics and the prepared experimental design. No legacy route record passes the complete current map-admission contract, and we claim no fastest route, crossover, or scaling law.

Section 2 positions this scope against the closest software and methodological work. Sections 3 and 4 state the mathematical and distributed contracts. Section 5 defines the retained problems, and Section 6 presents independent verification designs and diagnostic executions. The diagnostically admitted checks and their limitations appear in Section 7; Sections 8 and 9 summarize the resulting claim boundary.

2 Related Work

The closest literature separates into finite-element automation, differentiable mechanics, constitutive differentiation, and sparse derivative recovery. We organize it by the location of differentiation because software feature inventories do not determine whether two routes represent the same discrete functional.

The FEniCS project established symbolic weak-form specification and generated finite-element kernels [1]. DOLFINx extends this model with data-oriented parallel interfaces [2]. Automated adjoint systems operate at a higher program level: `dolfin-adjoint` and `pyadjoint` derive sensitivities of finite-element programs [11, 12], while `cashocs` provides adjoint-based optimal-control and shape-optimization workflows [13]. These systems motivate automation, but the present work concerns primal residual and tangent construction inside a Newton solve.

JAX provides composable program transformations for differentiation, vectorization, and compilation [14, 15]. JAX-FEM applies them to forward and inverse finite-element mechanics [3], and second-order implicit differentiation extends this setting to Hessian-vector products for differentiable physics [4]. AutoPDEx and JAX-CPFEM illustrate related energy-based and constitutive applications [16, 17]. These results establish the viability of differentiable finite elements; they do not compare the three derivative locations studied here under one distributed algebraic contract.

Bridge architectures are particularly close. Firedrake–JAX differentiates PDE solves through tangent and adjoint equations [5]. FEniCSx external operators admit general quadrature-point programs, including a JAX-differentiated Mohr–Coulomb model with Taylor tests [6]. JetSCI combines JAX-local discretizations with PETSc-owned distributed solves [7]. Thus neither constitutive AD nor the JAX–PETSc architecture is new here. Our distinction is the fixed-functional equivalence contract and the fail-closed comparison of element, constitutive, and colored routes.

Recent systems narrow this distinction further. Locality-aware AD performs per-element forward differentiation and sparse Hessian assembly on GPUs [8]. The `tatva` framework applies AD to a global energy and uses matrix-free products or coloring to materialize sparse tangents [9]. Hill and Dalle combine automatic local or global sparsity detection, matrix coloring, and sparse differentiation [10]. These papers establish both local and global AD strategies. The present contribution is therefore limited to explicit branchwise equivalence assumptions, distributed finite-element ownership conditions, and a common-accuracy route-selection protocol on CPU/MPI systems.

Sparse recovery itself is classical. Curtis, Powell, and Reid introduced directional probing for sparse Jacobians [18]; Coleman and Moré related sparse Jacobian and Hessian estimation to graph coloring [19, 20]. PETSc supplies the distributed vectors, matrices, Krylov methods, and preconditioners used in our implementation [21, 22]. We do not claim these algorithms as new. We use them to expose the assumptions needed to turn local Hessian actions into a canonical distributed matrix and to prevent timing comparisons from preceding derivative and endpoint agreement.

The branch-structured benchmark is motivated by Mohr–Coulomb slope stability, but its evidentiary role is intentionally narrower. Incremental plasticity, return mapping, and consistent linearization provide the continuum reference frame [23–25]. Variational strength reduction and recent three-dimensional continuation methods provide application context [26, 27]. Our implemented object is a synthetic endpoint potential with selected branches. It tests derivative placement; it does not validate a path-consistent constitutive history or a factor of safety.

3 Discrete Derivative Equivalence

We first define the algebraic object differentiated by every route. Let $V_{h,0} = \operatorname{span}\{\varphi_i\}_{i=1}^n$ be the constrained finite-element space, and let $x \in \mathbb{R}^n$ contain its free scalar coefficients. Essential data enter through an affine lift $\bar{u}_h(\theta)$, where θ collects fixed load and material parameters. The corresponding state is

$$u_h(x;\theta) = \bar{u}_h(\theta) + \sum_{i=1}^{n} x_i \varphi_i. \tag{1}$$

For element e, we write the local state as $u_e(x;\theta) = A_e x + b_e(\theta)$, where A_e is the fixed restriction and component map. This notation includes nonhomogeneous prescribed values and arbitrary free-DOF permutations.

Let B_{eq} map the local state to the strain or gradient argument at quadrature point q. We fix the quadrature rule, weights, local material data m_{eq}, and a deterministic first-match selector $\beta_{eq}(x)$ evaluated from the local strain and data. The discrete element contribution is

$$\Phi_e(x;\theta) := \sum_{q=1}^{n_q} w_{eq} \mathcal{W}_{\beta_{eq}(x)}\big(B_{eq}(A_e x + b_e); m_{eq}\big), \tag{2}$$

and the global free-space potential is

$$\Pi_h(x;\theta) := \sum_{e\in\mathcal{T}_h} \Phi_e(x;\theta) - f_h^\top x. \tag{3}$$

Additive terms that depend only on prescribed data may be omitted because they do not alter the free-space derivatives. The symbol $P_k(L_\ell)$ denotes degree-k Lagrange elements on mesh level L_ℓ.

3.1 Element and constitutive differentiation

Element AD applies first- and second-order differentiation directly to Φ_e. At a fixed state $x_\star$, write $\beta^\star_{eq} := \beta_{eq}(x_\star)$. Constitutive AD differentiates that selected scalar density with respect to

$$\varepsilon_{eq} := B_{eq}(A_e x + b_e), \qquad \sigma_{eq} := \partial_\varepsilon \mathcal{W}_{\beta^\star_{eq}}, \qquad C_{eq} := \partial^2_{\varepsilon\varepsilon} \mathcal{W}_{\beta^\star_{eq}}. \tag{4}$$

The branch-stability assumption below makes $\beta_{eq}(x) = \beta^\star_{eq}$ on a neighborhood of $x_\star$; the selector is therefore locally constant during branch-interior differentiation.

Proposition 1 (Fixed-functional branchwise equivalence)**.** *Fix an element state $x_\star$. Assume that, on a neighborhood of $x_\star$, (i) both routes use the same scalar densities, quadrature weights, material and history data; (ii) $u_e(x) = A_e x + b_e$ is affine; (iii) every B_{eq} is fixed; (iv) each selected density is twice continuously differentiable; (v) the branch labels and principal-value ordering do not change; and (vi) both routes use the same tensor and engineering-shear conventions. Then element AD and constitutive AD give identical element residuals and Hessians in exact arithmetic:*

$$\begin{aligned}
\nabla_x \Phi_e(x) &= \sum_q w_{eq} A_e^\top B_{eq}^\top \sigma_{eq}, \\
\nabla_x^2 \Phi_e(x) &= \sum_q w_{eq} A_e^\top B_{eq}^\top C_{eq} B_{eq} A_e.
\end{aligned} \tag{5}$$

Consequently, assembling either set of element objects gives the same global free-space gradient and Hessian of (3).

Proof. The affine map has zero second derivative. Applying the chain rule first to $\varepsilon_{eq}(x)$ and then to every summand of (2) gives (5). Summation and the fixed free-space assembly map preserve the identities. □

Proposition 1 is conditional. It does not apply at a branch switch, under an unresolved principal-value ordering, or when the compared codes change quadrature or constraints. It also establishes the derivatives of the implemented scalar potential, not equivalence to an incremental return-mapping model.

3.2 Colored sparse recovery

Let $H := \nabla_x^2 \Pi_h(x)$, let I_r be the free rows owned by rank r, and let $\mathcal{P}_r$ contain the structural entries (i,j) of those rows. The local element set $\mathcal{E}_r$ contains every element incident on I_r, and J_r contains every free column appearing in those elements. We write $H^{(r)}$ for the Hessian of the sum over $\mathcal{E}_r$, restricted to J_r. Because all element contributions incident on an owned row are present, $H^{(r)}_{ij} = H_{ij}$ for $(i,j) \in \mathcal{P}_r$.

The rank-local column-interference graph joins columns $j,k \in J_r$ whenever some owned row $i \in I_r$ satisfies $(i,j),(i,k) \in \mathcal{P}_r$. A local color class $\mathcal{C}_{r,c}$ defines the overlap seed

$$v_{r,c} := \sum_{j\in\mathcal{C}_{r,c}} e_j, \qquad y_{r,c} := H^{(r)} v_{r,c}. \tag{6}$$

This construction is the direct Hessian analogue of classical sparse derivative probing [18, 20].

Proposition 2 (Owned-row colored recovery). *For every rank r, assume that (i) $\mathcal{P}_r$ contains every active nonzero (i,j) of each owned row; (ii) $\mathcal{E}_r$ contains every element contributing to those rows, with complete state, affine-lift, material, and selected-branch data on its overlap; (iii) J_r contains every column in $\mathcal{P}_r$, and each owned row contains at most one structural entry from any $\mathcal{C}_{r,c}$; (iv) $H^{(r)}v_{r,c}$ is evaluated exactly with the complete overlap seed; and (v) every global row has one PETSc owner. Then, only for $(i,j) \in \mathcal{P}_r$ with $j \in \mathcal{C}_{r,c}$, assigning $H_{ij} := (y_{r,c})_i$ recovers every owned structural entry exactly. The union of the owned rows is the canonical distributed matrix H.*

Proof. For an owned row i, the identity $(y_{r,c})_i = \sum_{j\in\mathcal{C}_{r,c}} H^{(r)}_{ij}$ contains at most one structural nonzero by assumption (iii). Assumptions (i), (ii), and (iv) make the local sum complete and give $H^{(r)}_{ij} = H_{ij}$ on the owned pattern. The restricted assignment therefore recovers that entry, and unique ownership assembles each recovered row once. ☐

The proposition permits different rank-local color labels and counts because each rank evaluates its own complete incident-element Hessian and inserts only its owned pattern. It does not permit a missing incident element or structural edge, incomplete state or seed overlap, inconsistent branch data, or two same-color columns that meet an owned row. Our production recovery uses AD Hessian-vector products; a finite-difference gradient probe would add its truncation error.

3.3 Cost components and comparison variables

Let m_e be the active element DOF count, s the constitutive dimension, n_q the quadrature-point count, $n^{\mathrm{loc}}_{e,r}$ the overlap-element count on rank r, and c_r its local color count. Because the measured response is an MPI collective maximum, the structural work proxy must describe the busiest rank rather than the global number of uniquely owned elements. We therefore define

$$
\begin{aligned}
\chi_{\mathrm{elem}} &:= \max_r n^{\mathrm{loc}}_{e,r} n_q m_e^2,\\
\chi_{\mathrm{color}} &:= \max_r \bigl(n^{\mathrm{loc}}_{e,r} c_r\bigr) n_q m_e,\\
\chi_{\mathrm{const}} &:= \max_r n^{\mathrm{loc}}_{e,r} n_q \bigl(s^2 + s^2 m_e + s m_e^2\bigr).
\end{aligned} \tag{7}
$$

The first two expressions count the dense element-Hessian contribution shape and the vector work propagated by the exact AD-HVP seeds, respectively. In the constitutive expression, s^2 counts the dense tangent C_q, while $s^2 m_e$ and $s m_e^2$ count the two contractions used to form $B_q^\top C_q B_q$. The proxies in (7) are structural operation-shape counts, not exact FLOP, cache, or compiler-cost models. We therefore also decompose the measured route cost as

$$
T = T_{\mathrm{differentiate}} + T_{\mathrm{contract}} + T_{\mathrm{insert}} + T_{\mathrm{communicate}} + T_{\mathrm{cache}} + T_{\mathrm{imbalance}}, \tag{8}
$$

and record matrix nonzeros, overlap size, memory, branch fractions, rank count, and the applicable χ from (7). A separate one-factor-at-a-time diagnostic varies element DOFs, quadrature points, constitutive dimension, color count, nonzeros per row, message bytes, and imbalance. For each of four stages, an OLS model fits the log warm median on ranks 1 and 8 using an intercept, log rank, and the seven log factor ratios; rank 32 is used once for validation. Three independent allocation blocks are required per rank. Total synthetic work is held fixed while multi-rank imbalance is varied; nonunit imbalance is inapplicable at one rank and those two rows are excluded from the fit. The diagnostic must have median and 90th-percentile absolute percentage errors at most 50% and 100%, respectively. Because its kernels are synthetic rather than route-faithful, it is used only for descriptive mechanism checks and is not a production-model feature.

The final response is the logarithm of the collective-max warm-median tangent-construction time. Its 13 features are three route indicators, three route-specific cluster-architecture shifts, $\log(1+\chi)$, $\log(1+n^{\max}_{\mathrm{nz}})$, $\log(1+n^{\max}_{\mathrm{ov}})$, log rank, plastic fraction, and two untransformed max-to-mean imbalance ratios. More precisely, let $\mathcal{A} := \{\mathrm{elem}, \mathrm{color}, \mathrm{const}\}$, let I_a indicate route a, and let K indicate the cluster architecture. Furthermore, let $n^{\max}_{\mathrm{nz}}$ be the maximum number of owned matrix nonzeros over ranks, $n^{\max}_{\mathrm{ov}}$ the maximum overlap-DOF count, R the MPI rank count, p_{pl} the plastic quadrature fraction, and ι_e, ι_o the element and overlap max-to-mean ratios. The frozen model is

$$
\begin{aligned}
\log T = {}& \sum_{a\in\mathcal{A}} \alpha_a I_a + \sum_{a\in\mathcal{A}} \kappa_a K I_a + \beta_\chi \log(1+\chi_a) + \beta_{\mathrm{nz}} \log(1+n^{\max}_{\mathrm{nz}})\\
&+ \beta_{\mathrm{ov}} \log(1+n^{\max}_{\mathrm{ov}}) + \beta_R \log R + \beta_p p_{\mathrm{pl}} + \beta_e \iota_e + \beta_o \iota_o.
\end{aligned} \tag{9}
$$

The production OLS fit has no intercept: the three mutually exclusive route indicators provide the route-specific levels without intercept collinearity. Each active route–configuration–state–rank slot contributes one equally weighted row after its independent block medians are aggregated; censored or incomplete active-route groups are excluded and never imputed. Peak RSS and tracked allocation are outcomes, not predictors. Local single-node runs and cluster ranks 1 and 8 form the training set; cluster rank 32 is the untouched production holdout. Fitting requires at least 48 training rows, 20 holdout rows, 6/4 training/holdout rows per route, and 12/8 distinct training/holdout groups, with both architectures represented in training. The 13-column design must have full column rank and condition number at most 10^{10}. Percentage errors are computed in seconds after exponentiating the log-time predictions. The selector requires holdout median and 90th-percentile absolute percentage errors at most 25% and 50%, at least 90% correct ordering outside a 10% tie band, at least four resolved groups, and at least two distinct observed winners. A predictive selector is admissible only after these and all derivative, state, and endpoint gates pass. Coefficient, prediction, and route-order intervals use 10,000 paired allocation-block bootstrap resamples at 95% confidence with the prespecified seed 20260710. Section 7 shows that the current data do not meet this condition.

3.4 Mesh-aware nonlinear accuracy

Coefficient norms change with mesh density and therefore do not define a portable stopping contract. Let M_h be a symmetric positive-definite Riesz operator on the constrained free space. We use the primal and dual norms

$$\|s\|_{M_h} := (s^\top M_h s)^{1/2}, \qquad \|g\|_{M_h^{-1}} := (g^\top M_h^{-1} g)^{1/2}. \tag{10}$$

The scalar problems use a row-sum lumped P1 mass operator. Hyperelasticity and the branch-structured mechanics problem use the tangent of a reference elastic energy on the identical constrained free space. Before a reference tangent is admitted, a scale-aware symmetry check and a recorded direct factorization report zero negative and zero numerical zero pivots under the stated solver options. This is a numerical inertia check, not an analytic SPD proof. Each inverse-Riesz solve also passes an independently recomputed true-residual gate.

For nonlinear iteration $k \geq 1$, let

$$R_k := \|g_k\|_{M_h^{-1}}, \qquad C_k := \|x_k - x_{k-1}\|_{M_h}, \qquad S_k := \|x_k\|_{M_h}, \qquad D_k := |\mathcal{E}_h(x_k) - \mathcal{E}_h(x_{k-1})|. \tag{11}$$

We report the dimensionless quantities

$$r_k^{\mathrm{rel}} := \frac{R_k}{R_0}, \qquad c_k^{\mathrm{rel}} := \frac{C_k}{\max(s_0, S_k)}, \tag{12}$$

where r_k^{rel} is defined only when $R_0 > 0$. The scale $s_0 > 0$ has the same units as the primal norm and is either the initial state norm or an explicit physical scale. With absolute and relative tolerances denoted by τ_R^{abs}, τ_R^{rel}, τ_C^{abs}, and τ_C^{rel}, the maintained scalar, hyperelastic, and branch-structured Newton drivers accept an updated iterate only when

$$\begin{aligned} T_k = \{&x_k \text{ accepted and all terminal quantities finite}\} \\ &\wedge \{D_k < \tau_E\} \wedge \{C_k < \tau_C^{\mathrm{abs}} \ \vee \ c_k^{\mathrm{rel}} < \tau_C^{\mathrm{rel}}\} \\ &\wedge \{R_k < \max(\tau_R^{\mathrm{abs}}, \tau_R^{\mathrm{rel}} R_0)\}. \end{aligned} \tag{13}$$

Thus the energy-change test is required; it is not a reporting-only diagnostic. If $R_0 = 0$ exactly, the initial-relative residual is undefined and is not used; any finite implementation fallback value is treated as inapplicable. A zero-step stationary return then requires finite $\mathcal{E}_h(x_0)$ and $R_0 < \tau_R^{\mathrm{abs}}$; the energy-change and correction tests are inapplicable because no update has been accepted.

The Boolean logic in (13) is applied once for each scalar solve and each hyperelastic load step, while the branch-structured full solve applies it to its single prescribed load. A multi-step hyperelastic trajectory is complete only when every requested step is successful. The branch-structured publication gate additionally recomputes the endpoint residual and requires its selected-branch and provenance checks; these admission tests do not replace T_k. For hyperelasticity, the coefficient vector represents the free part of the deformation map under one fixed affine lift. Its reference-energy norm is used only within that representation and is not invariant under a change of lift. The endpoint record contains the selected Riesz operator, scale, free-space identity, numerical inertia record, inverse-solve diagnostics, and independently recomputed terminal norms.

The tolerances in the present diagnostic records are case-specific. The scalar Ginzburg–Landau pair supplies one local tenfold sensitivity check, but cross-mesh calibration remains open. The reference-elastic hyperelastic and branch-structured policies have passed operator smokes only; we therefore do not claim a universal or publication-calibrated stopping threshold.

3.5 Evidence and timing contract

Correctness precedes time. A route row first passes the common state hash, energy, gradient, matrix or multiple action probes, branch margin, Riesz residual, and correction gates. Timing then uses warmups, balanced route order, collective-max MPI phases, independent repetitions, and explicit censoring. Compilation, setup, assembly, preconditioner construction, Krylov solution, and total time remain separate quantities. A capped or failed solve is not encoded as a large successful time.

This hierarchy yields four distinct statements: a proved identity under stated assumptions; verification on finite tested states; a descriptive finite route map; and an empirical crossover or selector. Evidence for one level does not imply the next. In particular, near-equal energy does not replace a residual or tangent comparison, and one tangent probe does not establish complete matrix equality.

4 Distributed Construction and Numerical Contract

The implementation assigns local differentiable kernels to JAX and distributed algebraic objects to PETSc. Each MPI rank stores one contiguous interval of the free coefficient vector and the ghost values required by its overlap elements. The same ownership map is used for the residual, tangent rows, Riesz operator, and canonical state export. This common map is essential: comparing local arrays before undoing a route-specific permutation would not compare the same algebraic object.

4.1 Local derivative routes

The element route evaluates a scalar Φ_e and obtains its gradient and Hessian by JAX transformations. The constitutive route evaluates the same selected quadrature density, obtains σ_{eq} and C_{eq} from (4), and contracts them according to (5). For the plasticity experiments, the residual is also evaluated by element AD; only the tangent differentiation boundary changes. This choice prevents a residual-formula difference from being hidden inside a tangent comparison.

The colored route first builds the complete structural pattern of every owned row. It colors the overlap-local column-interference graph, forms one seed per color, evaluates JAX Hessian-vector products, and extracts the uniquely identified entries. The implementation does not assume that different ranks obtain the same color labels. It relies instead on Proposition 2 and unique PETSc row ownership.

All three routes use 64-bit arithmetic. Compilation and kernel warmup occur before measured repetitions. Compiled element, constitutive, and action kernels remain separate because their traces and intermediate arrays are different scientific costs.

4.2 Rank-local mesh and sparse assembly

Let I_r denote rank r's owned free DOFs and G_r its required ghosts. The retained overlap is

$$\mathcal{T}_r^{\mathrm{loc}} := \{e : \mathrm{dofs}(e) \cap I_r \neq \varnothing\}. \tag{14}$$

Before each local evaluation, a point-to-point exchange populates G_r and the affine lift supplies constrained values. Residual and tangent contributions are accumulated only for rows in I_r; off-rank columns retain global indices. PETSc's distributed coordinate format is preallocated once and reused.

Scalar energy requires a separate ownership rule because overlap elements occur on more than one rank. Each element has a deterministic owner $o(e)$ and weight $\omega_{r,e} = 1$ if $o(e) = r$. Hence

$$\sum_r \sum_{e \in \mathcal{T}_r^{\mathrm{loc}}} \omega_{r,e} \Phi_e = \sum_{e \in \mathcal{T}_h} \Phi_e. \tag{15}$$

This identity, the affine lift, the canonical free-DOF order, and the CSR indices are included in the distribution-equivalence test.

The production hyperelastic element path builds the structured mesh and overlap data rank locally. Its one-rank replicated construction is retained only as an algebraic comparator. The plasticity meshes and material data are immutable inputs; quadrature rules have stable names and are recorded in every result.

4.3 Nonlinear and linear solvers

The nonlinear driver accepts either line-search Newton or a trust-region policy [28, 29]. Armijo backtracking is used for the controlled Newton comparisons. The reduced-subspace trust method minimizes the quadratic model on the span of the Newton and gradient directions and applies the same configured Armijo rule when a post-step search is requested. Thus a method label fixes the actual globalization algorithm rather than selecting an undocumented fallback.

Each accepted Newton direction is produced by a PETSc Krylov solve on the current tangent. The result record contains the requested and effective KSP tolerances, iteration count, convergence reason, and independently recomputed true residual. A nonconverged KSP, nonfinite direction, rejected line search, trust-region rejection cap, or nonlinear iteration cap remains a failure. Such a row is never converted into a successful observation with a large time.

The primary verification configurations use ordinary Krylov methods and Hypre or direct MUMPS factors. Multigrid policies remain available for subsequent scaling experiments, but no preconditioner ranking is inferred in this paper. Reference-elastic Riesz operators are independent of the Newton tangent and remain fixed over the associated nonlinear trajectory.

4.4 Riesz operators and terminal recomputation

For a lumped scalar mass operator, the diagonal weights are assembled on the exact free space and must be finite and strictly positive. For mechanics, K_{ref} is copied from the exact discrete elastic tangent at the declared reference state. Hyperelasticity uses the identity deformation $y(X) = X$ with both end faces constrained. The Plasticity3D operator uses the elastic constitutive branch at zero displacement with the glued-bottom constraint.

The constrained matrix is first checked for scale-aware symmetry. A MUMPS factorization then reports its numerical inertia under the stored factorization and scaling options; admission requires

$$(n_-, n_0, n_+) = (0, 0, n_{\mathrm{free}}). \tag{16}$$

The dual norm is evaluated by an independent CG solve with a recomputed true- residual threshold. PETSc options cannot silently modify this solve unless an explicit, namespaced override is recorded. At every terminal state, the driver re-evaluates the energy, gradient, primal state norm, dual residual, and correction from the final stored vector. A failed run therefore cannot inherit an initial or pre-step residual as its endpoint value.

4.5 Canonical records and timing

Every runner writes strict JSON: nonfinite sentinels are serialized as `null`, and the file becomes visible only after an atomic replacement. The schema records the experiment, case, route, repetition, MPI size, threads, input and code identities, solver controls, terminal status, norms, work counters, timing phases, and state hashes. Canonical NPZ states use the original global ordering and include coordinate, connectivity, constraint, and metric metadata.

Fixed-state tangent construction is measured after one in-process warmup and five warm repetitions. Three-route comparisons use three independent cyclic blocks; the two-route high-order comparison uses four alternating blocks. The separate Tier-B full-solve confirmation instead uses ten independent cold-process paired blocks with no embedded warmup. Both series reduce elapsed time by the maximum over participating ranks and retain every rank-local value. The base route order is seeded by the first 16 hexadecimal digits of the SHA-256 of the comparison identifier, then cyclically rotated for three routes or alternated with its reverse for two. Setup, differentiation, contraction, insertion, communication, preconditioner, Krylov, and total phases are reported separately. The implementation also supports structural censoring when a route exceeds a prespecified memory or color-count bound. The active route set at a comparison point excludes only censors declared in the frozen contract. The high-order colored slot remains visible as a prespecified, risk-based non-attempt without a measured threshold; it is never called an out-of-memory observation, imputed, or used as a model row.

The local data available in this revision predate the complete timing design. They are diagnostically admitted only to correctness and covariate checks. The prepared distributed design uses the stricter paired-block implementation and has not contributed performance numbers to the manuscript.

5 Verification Problems and Test Design

The experiment hierarchy uses four problem blocks. Smooth scalar manufactured solutions measure spatial consistency. Hyperelasticity supplies analytic stress/tangent checks, nonaffine convergence, and a canonical MPI assembly test. A synthetic Mohr–Coulomb endpoint potential exercises constitutive branches and high-order quadrature. Controlled Ginzburg–Landau runs probe an indefinite Hessian and globalization failure semantics. Topology optimization and the two-dimensional plasticity surrogate are excluded from the central evidence because they do not satisfy the optimization and validation gates of this revision.

5.1 Manufactured p-Laplace problem

Let $p = 3$ and $\Omega = (0,1)^2$; the weak formulation follows the standard stationary p-Laplace setting [30, Ch. 2]. We consider

$$\mathcal{E}(u) := \int_\Omega \frac{1}{p}|\nabla u|^p - f u \, \mathrm{d}x, \tag{17}$$

with exact Dirichlet data and source determined by

$$u_\star(x,y) := x + y + 0.1\sin(\pi x)\sin(\pi y), \qquad f := -\nabla\cdot\big(|\nabla u_\star|\nabla u_\star\big). \tag{18}$$

The gradient satisfies $|\nabla u_\star| \geq 1 - 0.1\pi > 0.685$, avoiding the degenerate zero-gradient Hessian during the consistency test. The independent verifier uses uniform P1 triangles, a degree-five seven-point load/error rule, and a consistent NumPy/SciPy Newton tangent. It does not call the production JAX or PETSc assembly.

On the nonempty affine Dirichlet space, the gradient term is coercive and its strictly monotone derivative is injective on free variations; hence the finite-dimensional energy has a unique discrete minimizer. The measured quantities are the weighted L^2 error, the H^1-seminorm error, the weak residual, the tangent symmetry defect, and the minimum element-gradient norm. The expected P1 orders are two and one.

5.2 Manufactured Ginzburg–Landau problem

The real scalar energy with a source is

$$\mathcal{G}(u) := \int_\Omega \frac{\epsilon}{2}|\nabla u|^2 + \frac{1}{4}(u^2-1)^2 - f u \, \mathrm{d}x, \qquad \epsilon = 0.04. \tag{19}$$

We select

$$u_\star(x,y) := 0.8 + 0.1\sin(\pi x)\sin(\pi y), \qquad f := -\epsilon\Delta u_\star + u_\star(u_\star^2 - 1), \tag{20}$$

and impose its trace on $\partial\Omega$. The solve uses the same symmetric three-point triangle rule as the production JAX functional, while the error is evaluated by the seven-point rule. The exact solution satisfies $u_\star \geq 0.8 > 1/\sqrt{3}$, and the recorded computed nodal values also remain above this curvature threshold. Thus the tested states, rather than the exact solution alone, establish the positive-curvature restriction. This design measures spatial consistency without conflating it with selection among competing double-well basins.

The separate globalization diagnostic uses the zero-source production-style functional and one fixed initial state. It compares Newton–Armijo with the reduced-subspace trust method under the same ordinary GMRES–Hypre contract. One instance is sufficient to verify failure reporting, but not a robustness probability or timing distribution.

5.3 Hyperelasticity

Each mechanics benchmark below uses an explicitly nondimensional computational system. Let $L_0 > 0$ and $S_0 > 0$ denote its reference length and stress (equivalently, energy-density) scales. We report X/L_0, y/L_0, W/S_0, stress-like material parameters divided by S_0, and body-force densities multiplied by L_0/S_0. The stored arrays contain these scaled quantities, so the computational data take $L_0 = S_0 = 1$; we make no SI-unit or physical-calibration claim.

Let $y : \Omega \to \mathbb{R}^3$ be the deformation map, $F := \nabla y$, and $J := \det F > 0$. We use the compressible neo-Hookean density

$$W(F) := C_1\big(\operatorname{tr}(F^\top F) - 3 - 2\log J\big) + D_1(J-1)^2, \tag{21}$$

where $C_1 = 0.5$ and $D_1 = 5$ for the independent affine and manufactured verification problems on $\Omega = (0,1)^3$. The distributed cantilever uses $\Omega = [0, 0.4] \times [-0.005, 0.005]^2$, $C_1 = 3.8462 \times 10^7$, and $D_1 = 8.3333 \times 10^7$ in the same scaled system; run records retain the full binary64 inputs. Its first Piola stress is $P := \partial W/\partial F$ [31, Chs. 4–6]. The density is smooth on $J > 0$ but nonconvex in the free coefficients.

The affine patch maps a unit cube by

$$y(X) = F_\star X + t, \qquad F_\star = \begin{bmatrix} 1.15 & 0.08 & 0 \\ 0.02 & 0.92 & 0.05 \\ 0 & 0.03 & 1.08 \end{bmatrix}, \qquad \det F_\star = 1.139187. \tag{22}$$

An independent analytic implementation assembles W, P, the material tangent, nodal internal forces, and constant boundary tractions. The production JAX energy is differentiated on the identical six-tetrahedron mesh. Translation modes and a superposed rigid rotation test objectivity and covariance.

The nonaffine manufactured deformation is

$$y_\star(X,Y,Z) := \big(X + 0.05\sin(\pi X)\sin(\pi Y)\sin(\pi Z), Y, Z\big). \tag{23}$$

Its trace is imposed on the complete boundary, and the analytic body force is $b = -\operatorname{Div} P(\nabla y_\star)$. The independent P1 verifier uses uniform cube subdivisions 4, 8, 16, and 24, assembles P and its consistent tangent, and applies a fourth-order Duffy-product load rule. Orders six and eight provide a load-quadrature refinement check. It reports displacement L^2, deformation-gradient, first-Piola, and minimum-determinant errors. Initialization by the identity deformation targets the intended local basin.

The distributed fixed-state test uses the production cantilever geometry at a prescribed small rotation. One- and two-rank constructions share the mesh, free-space map, affine lift, state, direction, exact element Hessian, and MUMPS linear solve. The compared canonical objects are coordinates, connectivity, constraints, state, residual, CSR matrix, tangent action, and correction. Time is recorded only as a diagnostic because rank count is the varied factor.

For subsequent nonlinear solves, the fixed Riesz map is the exact Hessian of (21) at $y(X) = X$ on the two-end constrained free space. Its numerical inertia record and inverse-solve true residual are part of the stopping record.

5.4 Branch-structured endpoint potential

The branch benchmark is not an incremental constitutive model. It sets previous history variables to zero and defines a scalar endpoint potential from the total engineering strain

$$\varepsilon := (\varepsilon_{xx}, \varepsilon_{yy}, \varepsilon_{zz}, \gamma_{xy}, \gamma_{yz}, \gamma_{xz})^\top. \tag{24}$$

Engineering shear is converted to tensor shear before the ordered principal strains $\varepsilon_1 \geq \varepsilon_2 \geq \varepsilon_3$ are computed. A $10^{-15}\operatorname{diag}(0,1,2)$ perturbation supplies a deterministic tie convention at repeated values; it is not a material parameter.

For each material row, Young's modulus $E > 0$ and Poisson ratio $-1 < \nu < 1/2$ define

$$\mu := \frac{E}{2(1+\nu)}, \qquad K := \frac{E}{3(1-2\nu)}, \qquad \lambda_{\mathrm{L}} := K - \frac{2\mu}{3} = \frac{E\nu}{(1+\nu)(1-2\nu)}. \tag{25}$$

Here c_0, E, μ, K, and λ_{L} are scaled by S_0, whereas the unit weight γ is scaled by S_0/L_0; ν and the material angles are dimensionless. After Davis–B strength reduction, let $\bar{c}_\lambda$ and $s_\lambda := \sin\phi_\lambda$ denote the reduced cohesion and friction parameters. The trial yield quantity is

$$f_{\mathrm{tr}} := 2\mu\big((1+s_\lambda)\varepsilon_1 - (1-s_\lambda)\varepsilon_3\big) + 2\lambda_{\mathrm{L}} s_\lambda \operatorname{tr}\varepsilon - \bar{c}_\lambda. \tag{26}$$

The selected scalar potential has five labels: elastic, shear, left edge, right edge, and apex. Each plastic branch subtracts the corresponding nonnegative quadratic return term from an elastic or edge-restricted quadratic energy. Appendix A gives the denominators, multipliers, energies, and elastic-first tie order used by the code.

The assembled benchmark is

$$\Pi_h^{\mathrm{MC}}(x;\lambda_{\mathrm{sr}}) := \sum_{e\in\mathcal{T}_h}\sum_{q=1}^{n_q} w_{eq}\Phi_{\mathrm{MC}}\big(B_{eq}A_e x; \bar{c}_{\lambda,eq}, s_{\lambda,eq}, \mu_{eq}, K_{eq}, \lambda_{\mathrm{L},eq}\big) - f_h^\top x. \tag{27}$$

The heterogeneous slope mesh contains four material regions, has bounding box $[-175, 30]\times[0, 60]\times[0, 86.6025403784]$ in scaled coordinates, and uses the glued-bottom free space. The principal assembled route, stopping, and discretization studies use $\lambda_{\mathrm{sr}} = 1.55$. The earlier local fixed-element derivative diagnostic instead uses $\lambda_{\mathrm{sr}} = 1.50$; the two configurations are identified separately and are never pooled. Named tetrahedral rules use 1, 11, and 24 points for P_1, P_2, and P_4, respectively; an enriched 125-point Duffy rule is the fixed-state reference. Because the integrand is branchwise nonpolynomial, these names specify the discrete functional rather than an exactness claim.

The material-point matrix contains one strict interior state for every label, two-sided probes of all adjacent interfaces, rotated copies, and repeated- principal-value cases. A normalized branch margin records distance from the selected switching predicate. Ordinary Hessian claims are restricted to strict interiors. The finite-element tests add fixed P1, P2, and P4 elements, a constructed P2 state containing all five labels, and canonical tangent-action comparisons between the three derivative routes.

This block tests branch programming, contractions, quadrature sensitivity, and distributed derivative placement. It does not establish generalized differentiability at switches, semismooth convergence, a path-consistent plastic history, or physical slope-stability validation [23–25].

5.5 Admission sequence

Every numerical block follows the same order:

1. analytic, finite-difference, or manufactured verification;
2. canonical state, residual, matrix, and action comparison;
3. Riesz-scaled stopping and endpoint recomputation;
4. repeated timing under a common allocation and solver policy.

Failure at one stage prevents promotion to the next. The current paper reaches the first two stages for the reported local and fixed-state problems. It does not promote the unexecuted distributed route matrix to a performance result.

6 Verification

Verification proceeds from independent mathematical checks to route and MPI comparisons. Each block names the object being compared and the norm used for its gate. None of the tests below is interpreted as timing evidence.

6.1 Normalization and admission gates

For two scalars, vectors, or matrices of the same physical type, route and analytic defects use

$$\delta(a,b) := \begin{cases} 0, & \|a\| = \|b\| = 0,\\ \dfrac{\|a-b\|}{\max(\|a\|,\|b\|)}, & \text{otherwise}, \end{cases} \qquad \delta_{\mathrm{sym}}(A) := \begin{cases} 0, & \|A\|_{\mathrm{F}} = 0,\\ \dfrac{\|A-A^\top\|_{\mathrm{F}}}{\|A\|_{\mathrm{F}}}, & \text{otherwise}. \end{cases} \tag{28}$$

Here $\|\cdot\|$ denotes absolute value for scalars, the Euclidean norm for vectors, and the Frobenius norm for matrices. Every comparison involving a zero object is also reported in absolute terms. The distribution diagnostic instead uses the pilot-specific numerical floor $\max(1, \|a\|, \|b\|)$ in the stored units; this arbitrary absolute scale is one reason it is not reused as a physical endpoint norm.

Manufactured Newton solves report $\|R_{h,\mathrm{free}}^{(k)}\|_2/\|R_{h,\mathrm{free}}^{(0)}\|_2$ and require it to be at most 10^{-8}. Spatial errors are quadrature approximations of $\|u_h - u_\star\|_{L^2}$ and $\|\nabla u_h - \nabla u_\star\|_{L^2}$; the mechanics block additionally uses $\|P_h - P_\star\|_{L^2}$.

For a unit direction d, the centered action check is

$$D_h g(x;d) := \frac{g(x+hd) - g(x-hd)}{2h}, \qquad \delta\big(D_h g(x;d), H(x)d\big). \tag{29}$$

The smooth-energy gate uses $h = 3 \times 10^{-5}$; the fixed-element plasticity gate uses $h = 10^{-7}$; and the material-point gate uses $h = 10^{-6}$. The corresponding action tolerance is 10^{-7}, route defects must not exceed 10^{-8}, and smooth-branch symmetry defects must not exceed 10^{-10}. Complete step sweeps are retained rather than fitting the roundoff tail.

For the branch diagnostic, every predicate is divided by a scale of the same units: yield predicates use the maximum of the cohesion, trial value, and elastic stress terms, while multiplier/interface predicates use the maximum of the principal strains, return multipliers, return tests, and $\bar{c}_\lambda / \max(|\mu|, |K|, |\lambda_{\mathrm{L}}|)$. The active margin is the minimum signed normalized inequality required by the selected first-match row. Strict interiors require a margin of at least 10^{-3}; the separate quadrature exclusion band is 10^{-8}.

6.2 Manufactured and analytic checks

Table 1 summarizes the finest-pair rates or maximum analytic defect. The scalar problems use independent NumPy/SciPy assembly. The affine hyperelastic row compares the production JAX energy with independent analytic stress, tangent, and traction formulas. The nonaffine row is an independent manufactured solve.

Table 1: Independent manufactured and analytic verification. Orders use the finest mesh pair. The affine-patch defect is the maximum relative discrepancy among energy, residual, and Hessian.

Verification problem	L^2 order	H^1 order	Residual or defect	Scope
p-Laplace manufactured	1.999	0.999	8.90×10^{-9}	smooth P1 weak problem
Ginzburg–Landau manufactured	2.001	1.000	2.38×10^{-13}	controlled positive branch
Hyperelastic affine patch	–	–	4.58×10^{-15}	analytic W, P, and tangent
Hyperelastic nonaffine manufactured	1.887	1.006	3.11×10^{-13}	Piola order 0.983

For p-Laplace, all four levels reach the 10^{-8} relative weak-residual gate in three damped Newton steps. The successive L^2 orders are 1.986, 1.996, and 1.999; the H^1-seminorm orders are 0.990, 0.998, and 0.999. The finest errors are 2.76×10^{-5} and 5.45×10^{-3}. The minimum discrete gradient norm remains above 1.19, so the experiment avoids the zero-gradient degeneracy of the $p = 3$ Hessian.

The controlled Ginzburg–Landau branch converges in four Newton steps on every level. Its finest-pair orders are 2.001 and 1.000, with all nodal values between 0.8 and 0.901. These results verify the selected smooth branch and the production-style three-point quadrature. They do not establish basin robustness for the zero-source double-well problem.

6.3 Hyperelastic verification

On the affine patch, the relative energy, residual, and Hessian discrepancies are 4.58×10^{-15}, 2.67×10^{-15}, and 4.38×10^{-16}. Boundary-traction balance is 1.75×10^{-16} relatively. The net internal-force norm is 1.92×10^{-16} and the largest rigid-translation Hessian-action norm is 1.19×10^{-15}, both absolute values in the nondimensional force units of the unit-cube test. A superposed rotation gives zero stored energy defect and 7.94×10^{-17} Piola covariance error.

The nonaffine manufactured problem uses 4, 8, 16, and 24 subdivisions per axis. All four meshes converge in four damped Newton steps. The last-pair orders are 1.887 for displacement L^2, 1.006 for the deformation-gradient error, and 0.983 for first-Piola stress. The minimum computed determinant is 0.844, and tangent symmetry defects remain below 8.6×10^{-17}. Orders 4, 6, and 8 are compared for the analytic load. The largest induced response change is 8.29×10^{-6} of the finite-element error and the largest load change is 1.78×10^{-5} of the exact-interpolant consistency residual; both ratios decrease under refinement. Thus the test shows observed convergence for a nontrivial orientation-preserving formulation, but not the production rotating-boundary trajectory.

6.4 Derivative and branch checks

Table 2 reports the largest defect within each block. The route column compares analytic contractions or common-route objects. The finite-difference column evaluates centered directional checks at the prespecified step. A finite-difference error near 10^{-9} is consistent with roundoff–truncation balance and is not expected to match an AD identity at machine precision.

The smooth local block contains linear, nonlinear, indefinite, and admissible finite-strain energies. Across five cases, the maximum analytic gradient and Hessian discrepancies are 1.29×10^{-15} and 2.23×10^{-16}. The P1, P2, and P4 plasticity blocks each contain five fixed-element states. Their maximum element-versus-constitutive Hessian defect is 2.35×10^{-16}, and every state remains on one branch over the finite-difference gate interval.

The material-point matrix contains one strict interior for each of the five labels, 15 two-sided interface pairs, 15 rotated interiors, and seven repeated- spectrum cases. All finiteness, symmetry, transcription, rotation, and directional gates pass. The smallest normalized active-branch margin among the interiors is 0.125. The closest two-sided probes show selected-Hessian changes of order one, confirming that Proposition 1 must not be extended across switches.

Table 2: Fixed-state derivative verification. Plasticity rows use strict branch-interior states. The one/two-rank row compares canonical distributed objects assembled from local element Hessians; it does not test distributed colored recovery.

Block	Element states	Element route/ symmetry defect	FD/action defect	Assembled CSR defect	Scope
Smooth local energies	5	2.23×10^{-16}	3.31×10^{-9}	–	fixed element
$P_1(L_1)$ plasticity	5	1.20×10^{-16}	9.25×10^{-15}	–	fixed element
$P_2(L_1)$ plasticity	5	1.66×10^{-16}	1.09×10^{-14}	–	fixed element
$P_4(L_1)$ plasticity	5	2.34×10^{-16}	1.60×10^{-14}	–	fixed element
Mohr–Coulomb material point	5	8.68×10^{-17}	1.54×10^{-9}	–	five branch interiors; switches excluded
Hyperelasticity, one/two ranks	2	0.00	2.24×10^{-16}	–	fixed state and canonical ordering

6.5 Canonical distributed equivalence

The hyperelastic one- and two-rank workers reproduce the coordinates, connectivity, boundary map, affine lift, state, direction, and CSR indices as bitwise-identical stored arrays. The stored residual and matrix differences are zero in FP64. Relative errors are 8.67×10^{-19} in energy, 2.24×10^{-16} in the matrix action, and 2.33×10^{-16} in the MUMPS correction. Independently recomputed linear residuals are below 1.9×10^{-15}. This verifies the varied rank partition for rank-local element-Hessian assembly at one fixed state; it does not test distributed colored recovery, nonlinear endpoints, or parallel time.

6.6 Plasticity scope

The Plasticity3D tests verify the scalar branch program and its derivatives, not a distinct constitutive implementation. The NumPy material-point code independently implements rotations, finite differences, and interface construction, but transcribes the same endpoint energy. We therefore call the result internal verification. Physical validation would require a matched incremental return map and a recognized boundary-value benchmark with common history, constraints, and material assumptions.

The same limitation applies to repeated spectra. The deterministic tie convention prevents nonfinite AD values and provides a reproducible selected branch. It does not make the ordinary Hessian unique at an exact eigenvalue degeneracy.

Evidence status. The displayed values are diagnostic results of the revision implementation. They are traceable to strict artifacts and code hashes, but must be reproduced from the clean, immutable release commit before journal submission. No matched DOLFINx result is used: the unavailable local binary stack is treated as a scoped comparator omission rather than as evidence.

7 Evidence Adjudication

This section adjudicates the experiments that depend on quadrature, nonlinear stopping, or solver policy. The purpose is to determine which conclusions may follow from the verified derivatives. We report no publication timing because the available rows do not satisfy the paired and repeated design of Section 3.5.

7.1 Fixed-state quadrature sensitivity

Table 3 evaluates one prescribed fixed state for each degree with its degree-associated rule and with a common 125-point Duffy rule. The residuals are raw free-coefficient ℓ^2 diagnostics and are not stopping metrics. Tangent actions use one deterministic unit direction shared by every rule.

Table 3: Fixed-state quadrature sensitivity. The degree-associated and reference rules evaluate the same prescribed coefficient vector. Energy and tangent differences are relative to the 125-point rule.

Space	Degree-rule n_q	Relative energy difference	Degree-rule residual	Reference-rule residual	Relative tangent-action difference
$P_1(L_1)$	1	0.00	4.56×10^{-3}	4.56×10^{-3}	6.46×10^{-16}
$P_2(L_1)$	11	1.84×10^{-8}	2.47×10^{-3}	7.84×10^{2}	2.08×10^{-2}
$P_4(L_1)$	24	5.10×10^{-6}	5.53×10^{-3}	2.57×10^{3}	4.35×10^{-2}

The P1 integrand is constant on each affine tetrahedron, and all reported objects agree to roundoff. The P2 fixed state is nearly unchanged in energy: the 11-point and 125-point values differ by 1.84×10^{-8} relatively. However, its enriched-rule residual is 7.84×10^{2}, and the tangent action changes by 2.08%. For P4, the corresponding values are 5.10×10^{-6}, 2.57×10^{3}, and 4.35%. One P2 reference-rule sample lies inside the 10^{-8} branch-margin exclusion band. Its action is therefore the derivative of the deterministically selected branch, not an ordinary Hessian at a smooth point.

These results show that energy agreement alone is insufficient for a derivative comparison. Because every state is prescribed rather than obtained from a nonlinear solve, this experiment neither measures endpoint stationarity nor establishes a quadrature policy. A separate high-order solved-endpoint study remains necessary before we draw conclusions about mesh, degree, or quadrature convergence.

7.2 Mesh-aware stopping diagnostics

The reconstructible two-rank scalar sensitivity pair uses the zero-source Ginzburg–Landau problem and compares a baseline policy with ten-times tighter linear and nonlinear tolerances. For the reduced-trust solve, the lumped-L^2 state difference is 3.81×10^{-7}, or 2.22×10^{-7} relative, and the energy difference is 1.20×10^{-13}. The tighter policy uses fewer Newton iterations but more total Krylov iterations. One deterministic instance supports only a local tolerance-sensitivity observation, not a work ranking. The attempted p-Laplace pair is excluded because its random initial free vector was neither seeded nor stored; its numerical difference is therefore not a controlled calibration result. Appendix A.5 gives the complete retained configuration.

The hyperelastic reference-Riesz smoke gives the same inertia $(0, 0, 2133)$ and state scale on one and two ranks. Its dual norms differ by 7.43×10^{-12} relatively, and both inverse solves pass the independent 10^{-8} true-residual gate. Because the smoke takes no Newton step, it verifies the operator, MPI provenance, and terminal serialization only. It does not calibrate a nonlinear endpoint tolerance. The large CG/Jacobi iteration counts also preclude an efficiency conclusion for that inverse-norm policy.

The earlier one-step Plasticity3D comparison explains why stopping precedes route time. Tightening the KSP relative tolerance from 10^{-2} to 10^{-8} reduces the element-versus-constitutive state discrepancy to 5.28×10^{-11} relative. Both rows remain capped after one Newton step, so the observation diagnoses inexact linear solves and cannot certify a final endpoint.

7.3 Controlled globalization

For the retained Ginzburg–Landau instance, the controlled two-rank tier holds the discrete problem, deterministic starting state, Hessian route, ordinary Krylov solver, preconditioner, KSP tolerance, stopping settings, and thread count fixed. It changes only Newton–Armijo versus the reduced-subspace trust model with Armijo acceptance. The Newton direction is rejected by all 40 Armijo trials at the first iteration. The solver correctly retains the initial state and reports line-search failure. The reduced-trust method reaches its criteria in 12 iterations. This is one controlled robustness observation, not a success probability. We exclude the p-Laplace pair because the two cold processes used unseeded, unstored random initial vectors.

Both methods execute two hyperelastic load steps from the same identity state at the first load. The second load is warm-started from each method's own first endpoint, so its starting vectors are not known to coincide. The final energies differ by 8.15×10^{-5} under intentionally loose smoke tolerances, and no canonical state was stored. Those rows fail the endpoint gate. No solve-time value from the globalization diagnostic is therefore used.

7.4 Finite route map and cost-model decision

The strict analysis first clusters rows by exact canonical state. At each comparison point, the expected active route set is the frozen route set minus contract-declared censors; censored slots remain visible and are never imputed. A group is admitted to the finite map only when all active routes are present, the declared tangent-action error is at most 10^{-8}, and timing provenance proves the required MPI collective maximum. The twelve available local records predate that timing contract and are rejected before map admission. Their derivative arrays remain covered by the separate fixed-state checks, but none supplies an admitted route-map row.

Table 4 gives the fail-closed outcome. In addition to the timing-provenance failure, every local record was generated from a nonimmutable revision state. Consequently, zero training and zero holdout rows are eligible for the publication model. The selector status is `not fitted`; no descriptive local median is used for a speedup or crossover conclusion.

The distributed design covers $P_1(L_1)$, $P_1(L_2)$, $P_2(L_1)$, and $P_4(L_1)$; elastic and mixed states; and 1, 8, and 32 ranks. It includes visible prespecified censors, numerically checked Riesz-stopped confirmations, factorized cost measurements, multiple action probes, and endpoint validation. All routes in a timing block execute inside one allocation with randomized order and collective-max phase measurements. The design has been prepared and tested in dry-run mode, but it has not been executed and therefore contributes no numerical performance evidence.

Table 4: Current evidence decision for route selection. The legacy rows fail the timing-provenance gate before finite-map or predictive-model admission.

Evidence block	Rows	Current decision	Required promotion evidence
Fixed-state route map	0/12	no diagnostic map	collective-max proof and clean records
Predictive cost selector	0 train; 0 holdout	not admitted	paired distributed design and held-out validation
Descriptive paired route timing	0 diagnostic rows	not admissible	equal-accuracy repeated cluster runs
Post-fit crossover location	0 confirmation rows	not evaluated	separate hash-bound confirmation study
Synthetic factor diagnostic	1 recorded failures	descriptive diagnostic reported	not a selector gate

7.5 Excluded optimization evidence

The topology implementation now distinguishes normalized material fraction from material measure. On the corrected 64×32 three-step diagnostic, one and two ranks agree within 4.60×10^{-6} relatively in compliance and 2.68×10^{-6} in weighted density. Both rows nevertheless miss the material target, every design subsolve reaches its iteration cap, and no KKT metric is available. The result is a distributed unit/parity check, not an optimization solution. Topology is therefore removed from the central paper.

Adjudication. The diagnostically admitted results support, on the tested states, the stated derivative identities, independent spatial consistency, canonical fixed-state distribution, and several negative claim boundaries. They do not establish a route ranking, mesh-independent nonlinear accuracy, high-order quadrature convergence, or parallel scaling.

8 Discussion

The analysis separates a mathematical identity from its computational realization. On a fixed smooth branch, element and constitutive AD are two applications of the chain rule. Colored recovery adds a second proposition: the structural pattern, interference coloring, overlap state, and row ownership must be compatible with every recovered entry. The fixed-element and canonical MPI tests do not provide one joint test of these claims: the fixed-element artifacts support element–constitutive equivalence, serial regression tests exercise colored assembly, and the reported one–two-rank MPI pilot uses the element-Hessian route. Thus distributed colored recovery remains an open experimental gate. None of these checks proves that an implementation remains correct after a branch switch, a change of quadrature, or an incomplete ghost exchange.

The quadrature diagnostic illustrates why the discrete functional belongs in the comparison contract. At the prescribed $P_2(L_1)$ and $P_4(L_1)$ states, the degree-associated energies differ from the enriched evaluation by only 1.84×10^{-8} and 5.10×10^{-6} relatively. Nevertheless, the corresponding tangent actions differ by 2.08% and 4.35%. Thus energy agreement alone cannot establish derivative equivalence or quadrature adequacy. Because these states are prescribed, the experiment does not assess stationarity or establish a degree trend. Moreover, one P2 reference sample lies inside the declared switch-exclusion band, so the reported action is a selected-branch sensitivity rather than an ordinary smooth Hessian action.

The branch tests have a similarly precise boundary. All five selected Mohr–Coulomb labels are exercised at strict interior points, and rotations, repeated spectra, and two-sided interface probes remain finite under the stated tie convention. The selected Hessian may change by order one across an interface. We therefore use ordinary gradients and Hessians only in branch interiors. No generalized derivative, semismooth convergence theorem, or incremental plastic-history validation follows from these tests.

The current route data do not yet determine an admitted finite route map or a performance model. Twelve local records contain useful exact-state and tangent-action diagnostics, but they predate the collective-max timing record and clean-provenance contracts. A valid crossover study must measure all routes within balanced allocations, use collective-max MPI timings, include independent block repetitions, and admit endpoints under the same Riesz-scaled stopping rule. The prepared protocol also separates quadrature, local differentiation, contraction, sparse insertion, communication, cache, memory, and imbalance costs. Until those measurements are executed, no fastest-route, speedup, selector, or scaling statement is justified.

The independent manufactured problems verify consistency rather than physical validity of the production benchmarks. The scalar solutions are smooth, and the recorded Ginzburg–Landau endpoints lie in a controlled positive branch. The nonaffine hyperelastic problem uses exact data on the complete boundary, while the affine patch supplies the reaction and objectivity checks. These results support the energy, residual, and tangent implementation, but they do not validate the rotating-beam trajectory or certify local minimality of a nonconvex endpoint.

The resulting scope is narrower than a general nonlinear finite-element framework comparison. It is also more reproducible: every diagnostically admitted claim names the differentiated functional, constrained space, branch status, norm, state identity, and distributed object. Future performance work should preserve this contract rather than trading accuracy for apparent route speed.

9 Conclusions

We formulated and tested a conditional equivalence framework for three placements of automatic differentiation in nonlinear finite elements. Element and constitutive differentiation agree when they act on the same quadrature-defined potential, affine constraint map, strain convention, and smooth active branch. Colored sparse recovery produces the same operator only under additional pattern, interference, overlap, and row-ownership conditions.

The prerelease diagnostic verification supports element–constitutive equivalence on smooth local energies and high-order branch-interior plasticity states. A separate canonical distributed hyperelastic check verifies rank-local element-Hessian assembly, while distributed colored recovery remains open. Independent manufactured solutions recover the expected P1 spatial behavior, including nonaffine hyperelastic displacement, deformation-gradient, and stress convergence. Conversely, the enriched-rule study rejects endpoint energy as a sufficient quadrature diagnostic.

These results must be reproduced from the clean release commit before submission. They diagnose derivative correctness on the tested states, not a universal computational ranking. The available timings fail the prespecified publication-model gates, so this revision makes no speedup, crossover, or scaling claim. The next empirical step is the prepared paired CPU/MPI design with common Riesz stopping, repeated within-allocation route blocks, canonical endpoint comparison, and factorized cost measurements. Its outcome may support a route selector; the mathematical equivalence and evidence hierarchy do not depend on that outcome.

A Branch and Solver Details

A.1 Complete endpoint-branch definition

This subsection records the exact scalar branch program used in Section 5.4. Let c_0, ϕ, and ψ denote the input cohesion, friction angle, and dilation angle. For strength-reduction factor $\lambda_{\mathrm{sr}} > 0$, the Davis–B map is defined as in the stated slope-stability reduction context [26, 32, 33]:

$$
\begin{aligned}
c_1 &:= \frac{c_0}{\lambda_{\mathrm{sr}}}, & \phi_1 &:= \arctan\left(\frac{\tan\phi}{\lambda_{\mathrm{sr}}}\right), & \psi_1 &:= \arctan\left(\frac{\tan\psi}{\lambda_{\mathrm{sr}}}\right), \\
\beta &:= \frac{\cos\phi_1\cos\psi_1}{1-\sin\phi_1\sin\psi_1}, & c_\lambda &:= \beta c_1, & \phi_\lambda &:= \arctan(\beta\tan\phi_1),
\end{aligned}
\tag{30}
$$

with

$$
\bar{c}_\lambda := 2c_\lambda\cos\phi_\lambda, \qquad s_\lambda := \sin\phi_\lambda. \tag{31}
$$

The tested material table has $\psi = 0$, but the implementation retains the complete map (30). The mathematical domain used below has finite ordered principal strains, $K > 0$, $\mu > 0$, $-1 < s_\lambda < 1$, and $s_\lambda \neq 0$ whenever the apex row is reachable. On this domain the displayed denominators are nonzero. Defensive tiny-denominator substitutions in the program are failure containment outside this model; an admitted row must instead pass the stored denominator-margin gate.

Let $I_1 := \operatorname{tr}\varepsilon$ and

$$
Q(\varepsilon) := \varepsilon_{xx}^2 + \varepsilon_{yy}^2 + \varepsilon_{zz}^2 + \frac{1}{2}\left(\gamma_{xy}^2 + \gamma_{yz}^2 + \gamma_{xz}^2\right). \tag{32}
$$

The elastic energy is

$$
\Phi_{\mathrm{el}} := \frac{1}{2}\lambda_{\mathrm{L}} I_1^2 + \mu Q(\varepsilon). \tag{33}
$$

In addition to f_{tr} from (26), the return tests are

$$
\begin{aligned}
r_{\mathrm{sl}} &:= \frac{\varepsilon_1-\varepsilon_2}{1+s_\lambda}, & r_{\mathrm{sr}} &:= \frac{\varepsilon_2-\varepsilon_3}{1-s_\lambda}, \\
r_{\mathrm{la}} &:= \frac{\varepsilon_1+\varepsilon_2-2\varepsilon_3}{3-s_\lambda}, & r_{\mathrm{ra}} &:= \frac{2\varepsilon_1-\varepsilon_2-\varepsilon_3}{3+s_\lambda}.
\end{aligned}
\tag{34}
$$

Their strictly positive denominators are

$$
\begin{aligned}
D_{\mathrm{s}} &:= 4\lambda_{\mathrm{L}} s_\lambda^2 + 4\mu(1+s_\lambda^2), \\
D_{\mathrm{l}} &:= 4\lambda_{\mathrm{L}} s_\lambda^2 + \mu(1+s_\lambda)^2 + 2\mu(1-s_\lambda)^2, \\
D_{\mathrm{r}} &:= 4\lambda_{\mathrm{L}} s_\lambda^2 + 2\mu(1+s_\lambda)^2 + \mu(1-s_\lambda)^2, \\
D_{\mathrm{a}} &:= 4K s_\lambda^2.
\end{aligned}
\tag{35}
$$

For the tested parameters $s_\lambda > 0$; the verifier records normalized denominator margins.

The four plastic multipliers are

$$\begin{aligned}
\Lambda_{\mathrm{s}} &:= \frac{f_{\mathrm{tr}}}{D_{\mathrm{s}}},\\
\Lambda_{\mathrm{l}} &:= \frac{\mu\big((1+s_\lambda)(\varepsilon_1+\varepsilon_2) - 2(1-s_\lambda)\varepsilon_3\big) + 2\lambda_{\mathrm{L}} s_\lambda I_1 - \bar{c}_\lambda}{D_{\mathrm{l}}},\\
\Lambda_{\mathrm{r}} &:= \frac{\mu\big(2(1+s_\lambda)\varepsilon_1 - (1-s_\lambda)(\varepsilon_2+\varepsilon_3)\big) + 2\lambda_{\mathrm{L}} s_\lambda I_1 - \bar{c}_\lambda}{D_{\mathrm{r}}},\\
\Lambda_{\mathrm{a}} &:= \frac{2K s_\lambda I_1 - \bar{c}_\lambda}{D_{\mathrm{a}}}.
\end{aligned} \tag{36}$$

The associated energies are

$$\begin{aligned}
\Phi_{\mathrm{s}} &:= \Phi_{\mathrm{el}} - \frac{1}{2} D_{\mathrm{s}} \Lambda_{\mathrm{s}}^2,\\
\Phi_{\mathrm{l}} &:= \frac{1}{2}\lambda_{\mathrm{L}} I_1^2 + \mu\left(\varepsilon_3^2 + \frac{1}{2}(\varepsilon_1+\varepsilon_2)^2\right) - \frac{1}{2} D_{\mathrm{l}} \Lambda_{\mathrm{l}}^2,\\
\Phi_{\mathrm{r}} &:= \frac{1}{2}\lambda_{\mathrm{L}} I_1^2 + \mu\left(\varepsilon_1^2 + \frac{1}{2}(\varepsilon_2+\varepsilon_3)^2\right) - \frac{1}{2} D_{\mathrm{r}} \Lambda_{\mathrm{r}}^2,\\
\Phi_{\mathrm{a}} &:= \frac{1}{2} K I_1^2 - \frac{1}{2} D_{\mathrm{a}} \Lambda_{\mathrm{a}}^2.
\end{aligned} \tag{37}$$

The elastic-first selection order is

$$\Phi_{\mathrm{MC}}(\varepsilon) := \begin{cases}
\Phi_{\mathrm{el}}, & f_{\mathrm{tr}} \le 0,\\
\Phi_{\mathrm{s}}, & f_{\mathrm{tr}} > 0 \text{ and } \Lambda_{\mathrm{s}} \le \min(r_{\mathrm{sl}}, r_{\mathrm{sr}}),\\
\Phi_{\mathrm{l}}, & f_{\mathrm{tr}} > 0 \text{ and } r_{\mathrm{sl}} < r_{\mathrm{sr}} \quad \text{and} \quad r_{\mathrm{sl}} \le \Lambda_{\mathrm{l}} \le r_{\mathrm{la}},\\
\Phi_{\mathrm{r}}, & f_{\mathrm{tr}} > 0 \text{ and } r_{\mathrm{sr}} < r_{\mathrm{sl}} \quad \text{and} \quad r_{\mathrm{sr}} \le \Lambda_{\mathrm{r}} \le r_{\mathrm{ra}},\\
\Phi_{\mathrm{a}}, & f_{\mathrm{tr}} > 0 \text{ and none of the three tests above holds,}
\end{cases} \quad . \tag{38}$$

The cases are evaluated from top to bottom and the first true row is returned; mutual exclusivity of the un-ordered plastic predicates is therefore not assumed. The predicates are part of the discrete surrogate. Differentiation treats the selected row of (38) as fixed.

A.2 Interface and repeated-spectrum probes

For every adjacent branch pair, the material-point verifier constructs states on both sides of the switching predicate. Each state records the normalized active-branch margin, constitutive denominator, principal-value gap, finite stress and tangent, tangent symmetry, and centered finite-difference action. The two one-sided selected Hessians are compared descriptively. Their difference does not define a generalized derivative.

Repeated principal values are evaluated under the deterministic diagonal perturbation stated in Section 5.4. The checks require finite energy, stress, and tangent, but they do not claim rotational invariance of the arbitrary tie-selected tangent at exact degeneracy. Separate rotations of strict interior states test energy invariance and stress/tangent covariance.

A.3 Stopping diagnostics

The retained scalar lumped-mass pilot compares baseline and ten-times tighter linear and nonlinear tolerances on the deterministic zero-source Ginzburg–Landau instance. Its relative final-state difference is 2.22×10^{-7}. This value is a local sensitivity observation; it neither calibrates a cross-mesh stopping threshold nor provides timing evidence. The attempted p-Laplace pair is excluded because its random initial coefficients were not seeded or stored.

The hyperelastic reference operator was checked independently on one and two ranks for a P1 problem with 2,133 free DOFs. Both matrices have inertia $(0, 0, 2133)$ and the same initial deformation-map norm 1031.6632492383628. The reported dual residuals differ by 7.43×10^{-12} relatively, and both inverse-Riesz solves satisfy the 10^{-8} true-residual gate. This is an operator and serialization test with zero permitted Newton iterations, not a nonlinear endpoint comparison. The large Jacobi-preconditioned CG counts also motivate a separate efficiency calibration before production-scale use.

A.4 Prepared distributed comparison

The executable design spans $P_1(L_1)$, $P_1(L_2)$, $P_2(L_1)$, and $P_4(L_1)$; elastic and mixed-branch states; and 1, 8, and 32 ranks. The high-order colored slot is retained as a prespecified risk-based non-attempt, not a measured resource failure. Full-solve confirmations use numerically checked reference-elastic stopping and a strict endpoint validator.

The design is intentionally absent from the numerical result tables because it has not been executed. The exact execution inventory and resource ceilings are kept in the archived experiment protocol rather than the manuscript. Three-route fixed-state comparisons use three cyclic blocks, the two-route high-order comparison uses four alternating blocks, and full-solve confirmations use ten paired cold-process blocks. The base order is obtained from the first 16 hexadecimal digits of the SHA-256 of the comparison identifier. The release gate additionally requires collective-max phase timing, multiple derivative probes, and a cross-route endpoint analyzer. These conditions prevent scheduler, node, cache, or stopping differences from being interpreted as derivative-route speed.

A.5 Diagnostic configurations

The scalar manufactured meshes have 8, 16, 32, and 64 uniform subdivisions per coordinate direction. The p-Laplace solve starts from the affine extension $x + y$ and the Ginzburg–Landau solve starts from its exact Dirichlet extension; both use damped Newton iteration and the relative residual defined in Section 6.1. The nonaffine hyperelastic meshes and material constants are stated in Section 5.3. The affine patch uses one unit cube divided into six tetrahedra.

The zero-source Ginzburg–Landau diagnostics instead use the checked-in level-5 P1 mesh of $[-1, 1]^2$, with 4,225 nodes, 8,192 triangles, and 3,969 free coefficients. They set $\epsilon = 0.01$, $f = 0$, and $u = 0$ on the complete boundary, and initialize the free coefficients by

$$u_0(x, y) := \sin\left(\frac{\pi(x-1)}{2}\right) \sin\left(\frac{\pi(y-1)}{2}\right). \tag{39}$$

The lumped-L^2 sensitivity records hash the mesh arrays and final states. They do not contain a digest of the initial vector; equation (39) and the hashed mesh arrays are therefore the available start-state identity.

The lumped-L^2 sensitivity pair uses two MPI ranks, one thread per rank, element Hessians, GMRES–Hypre, reduced-subspace trust with Armijo acceptance, and at most 100 Newton iterations. Define the tolerance vector by

$$\boldsymbol{\tau} := (\tau_E, \tau_R^{\text{abs}}, \tau_R^{\text{rel}}, \tau_C^{\text{abs}}, \tau_C^{\text{rel}}).$$

The two policies are

$$\boldsymbol{\tau}_{\text{base}} = (10^{-6}, 10^{-5}, 10^{-3}, 10^{-10}, 10^{-3}), \qquad \boldsymbol{\tau}_{\text{tight}} = (10^{-7}, 10^{-6}, 10^{-4}, 10^{-11}, 10^{-4}).$$

The KSP relative tolerance and cap change from $(10^{-3}, 200)$ to $(10^{-4}, 300)$. The common state scale is the unit-field lumped norm $s_0 = 1.96875$. The attempted production p-Laplace pair used independent calls to an unseeded random initializer and stored neither initial vector; we therefore report none of its numerical differences.

The distributed hyperelastic diagnostic uses the brick $[0, 0.4] \times [-0.005, 0.005]^2$, an $80 \times 2 \times 2$ cell grid split into 1,920 tetrahedra, 729 nodes, and 2,133 free vector DOFs after both end faces are constrained. Its constants are $C_1 = 3.8462 \times 10^7$ and $D_1 = 8.3333 \times 10^7$ in the nondimensional stress scale of Section 5.3; run records retain the full binary64 values. The stored fixed state twists the right face by 0.15 radians; no nonlinear trajectory is inferred from that state.

The Plasticity3D L_1 macro mesh has 3,845 vertices and 18,419 tetrahedra. Its bounding box is $[-175, 30] \times [0, 60] \times [0, 86.6025403784]$ in the scaled coordinates of Section 5.3. The P_1, P_2, and P_4 glued spaces have 10,526, 79,024, and 610,964 free vector DOFs. Source labels constrain x on groups 1 and 2, y on group 5, and z on groups 3 and 4; the maintained case additionally fixes every component on $y = 0$. Gravity acts in the negative y direction. The four material rows are listed below. Here c_0 and E are scaled by S_0, γ is scaled by S_0/L_0, and ϕ, ψ, and ν are dimensionless. The derived elastic moduli follow (25).

region	c_0	ϕ (deg)	ψ (deg)	E	ν	γ
cover layer	15	30	0	10000	0.33	19
general foundation	15	38	0	50000	0.30	22
weak foundation	10	35	0	50000	0.30	21
slope mass	18	32	0	20000	0.33	20

with identical saturated and unsaturated unit weights in the tested dry table. The source input is the checked-in `SSR_hetero_ada_L1.msh`; every publication run must record its content hash.

The local fixed-element derivative diagnostic uses $\lambda_{\text{sr}} = 1.50$. The principal assembled route, stopping, quadrature, and any conditional scaling protocols use $\lambda_{\text{sr}} = 1.55$. These configurations remain separate in every evidence record and comparison.

The controlled globalization diagnostic uses two MPI ranks and one thread per rank. Its retained Ginzburg–Landau pair uses the problem and deterministic initial state above, the coefficient ℓ^2 metric, GMRES–Hypre with relative tolerance 10^{-3} and cap 200, and the baseline tolerance tuple just stated, with a cap of 100 Newton iterations. The compared methods are Newton–Armijo and reduced-subspace trust with Armijo acceptance. The pilot stores no canonical endpoint state. The attempted p-Laplace pair is excluded because its two unseeded random initial vectors were not stored.

The hyperelastic pair uses the procedural level-2 P1 grid with $160 \times 4 \times 4$ brick cells, split into 15,360 tetrahedra, on $[0, 0.4] \times [-0.005, 0.005]^2$. It has 12,075 total and 11,925 free vector DOFs, $C_1 = 3.8461538461538464 \times 10^7$, and

$D_1 = 8.333333333333333 \times 10^7$. The left face retains the identity map and the right face rotates about the x-axis. The two steps prescribe angles $\pi/3$ and $2\pi/3$; the first starts from the identity map and the second warm-starts from the method's first endpoint. Both methods use coefficient ℓ^2 stopping, GMRES–GAMG with relative tolerance 10^{-1} and cap 30, at most 100 Newton iterations per load, and $(\tau_E, \tau_R^{\mathrm{abs}}, \tau_R^{\mathrm{rel}}, \tau_C^{\mathrm{abs}}, \tau_C^{\mathrm{rel}}) = (10^{-4}, 10^{-3}, 10^{-3}, 10^{-10}, 10^{-4})$. No initial or final state hash was stored for these smoke rows.

For every retained globalization row, Armijo uses initial step 1, sufficient-decrease constant 10^{-4}, contraction 1/2, and at most 40 trials. The reduced trust model uses initial, minimum, and maximum radii 1, 10^{-8}, and 10^6, shrink/expand factors 1/2 and 3/2, acceptance-update thresholds 0.05 and 0.75, and at most six rejected candidates per Newton iteration. Section 7.3 states why none of these rows supports a timing comparison.

The diagnostic environment recorded Python 3.12.10, NumPy 2.4.2, JAX/JAXlib 0.9.1, petsc4py 3.24.2, and Open MPI 5.0.10. These versions describe the dirty pilots only. A clean publication record must repeat the full compiler, BLAS, PETSc external-package, CPU, operating-system, placement, and environment manifest.

Code and Data Availability

The primary implementation, checked-in inputs, strict run records, canonical state exports, experiment protocols, and table-generation scripts are available in the development repository `https://github.com/Beremi/fenics_nonlinear_energies`. The present repository artifacts are prerelease diagnostics. A versioned archive containing admitted clean reruns and immutable manifests is required before submission; no archival DOI is claimed for the current evidence.

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